# 2024 Annual Water Quality Report

## PACE UNIVERSITY

Cover art: Leanna Machado '24

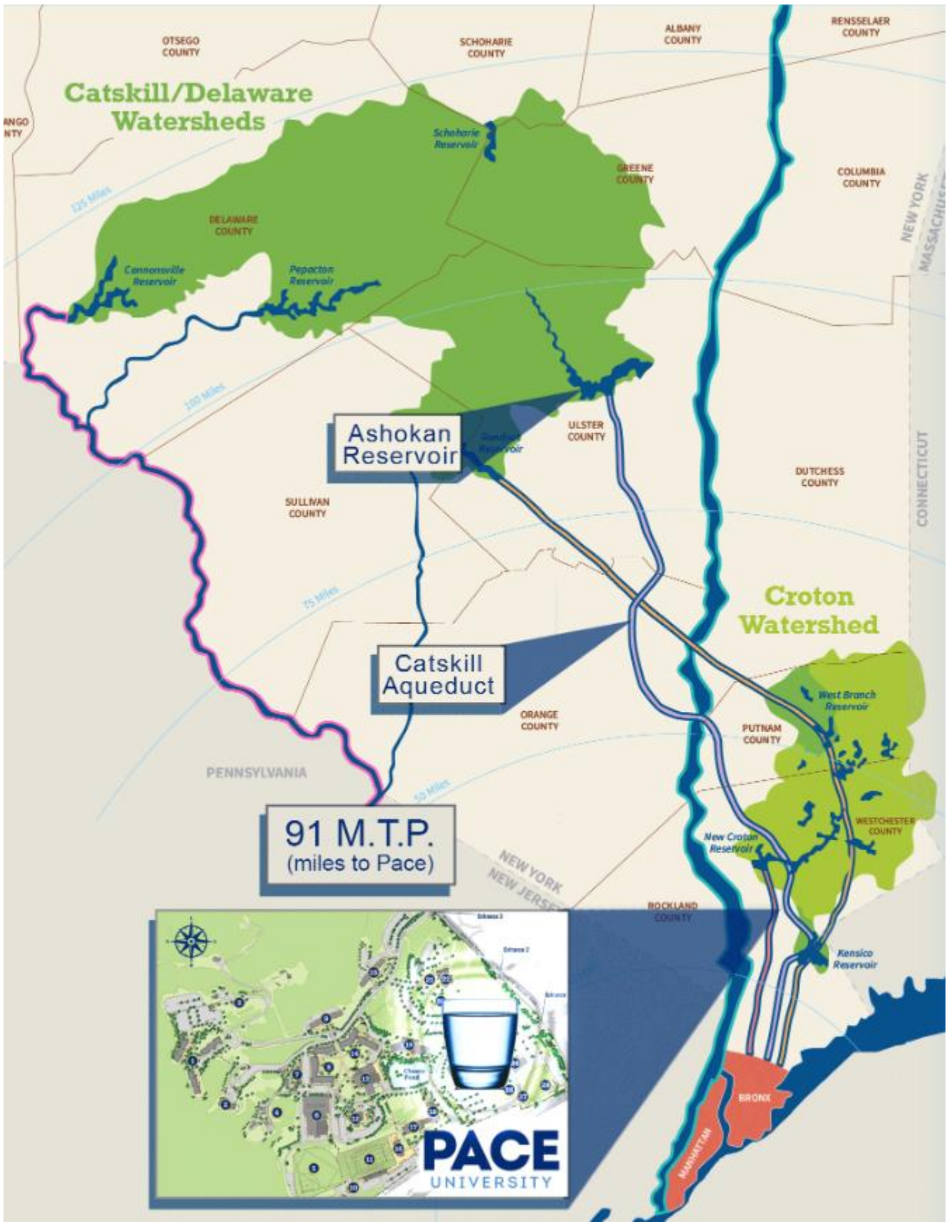
Gale Epstein Center Edition

## TABLE OF CONTENTS


| Section | Page # |
| --- | --- |
| Water Supply System Map | 3 |
| Introduction | 4 |
| Where does our water come from? | 5 |
| Treating our Drinking Water | 8 |
| Facts and Figures | 11 |
| Testing for Drinking Water Quality | 11 |
| Sampling and Monitoring | 11 |
| Regulation of Drinking Water | 12 |
| Conserving our Supply | 15 |
| What are regular precautions I should take? | 15 |
| Do's & Don'ts of Water Conservation | 15 |
| Information on Fluoride Addition | 17 |

| | |
|---|---|
| Information on Cryptosporidium and Giardia | 17 |
| Is there lead in my drinking water? | 19 |
| Is our water system meeting other rules that govern operations? | 20 |
| Are there contaminants in our drinking water? | 20 |
| Drinking Water Quality Tables | 21 |
| What does this information mean? | 23 |
| Lead | 23 |
| FAQ | 25 |
| Glossary | 28 |
| Closing Statement | 34 |
| Contact Information | 34 |

## WATER SUPPLY SYSTEM MAP



**INTRODUCTION**

To comply with State regulations, Pace University annually issues a report by May 31st describing the quality of your drinking water for the previous calendar year (January – December). The purpose of this report is to enable you, the consumer, to make informed decisions about your health as it relates to the consumption of water, and raise your awareness about the need to protect our drinking water.

**Last year, your tap water met all State drinking water health standards. We are proud to report that our system did not violate a maximum contaminant level or any other water quality standard.**

This report provides an overview of last year's water quality. Included are details about where your water comes from, what it contains, and how it compares to State standards. The report lists contaminants for which testing has been conducted, and separately lists whether they have exceeded state or federal standards. The State and the EPA enforce regulations that limit the amount of certain contaminants allowed in water delivered by public water systems.

If you have any questions about this report or concerning your drinking water, please contact the Pace University Facilities Management Office at (914) 923-2840. We want you to be informed about your drinking water. To obtain additional information on the drinking water provided to Pace University, you may contact the Mount Pleasant Water District directly, at 119 Lozza Drive, Valhalla, New York 10595-1268, or by telephone at (914) 742-2313. The source water assessment report and the Annual Drinking Water Quality Reports for 2024 prepared by the Mount Pleasant Water District and the New Castle Consolidated Water District are also available at the Pace University campus library. In addition, you can visit www.mynewcastle.org for a copy of the New Castle/Stanwood Consolidated Water System report.

## WHERE DOES OUR WATER COME FROM?

### WHERE OUR WATER COMES FROM

TIMELINE

**1905** — CATSKILL WATER SUPPLY DEVELOPMENT


CATSKILL AQUEDUCT

Initiated the construction of reservoirs, tunnels, and infrastructure to deliver clean and reliable water from the Catskill Mountains to New York City.

**1905** — WATER SUPPLY ACT OF 1905

The provision and state law that required New York City to allow local communities, along the path of the aqueduct, to connect to the water supply infrastructure


ASHOKAN RESERVOIR

**1915** — THE CATSKILL AQUEDUCT


ASHOKAN RESERVOIR

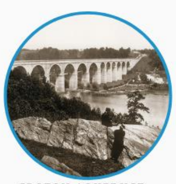
The first components of the Catskill System are completed: Ashokan, Kensico, and Hillview reservoirs, as well as the Catskill Aqueduct; City Tunnel #1 put was also into service.

**1930** — THE TOWN OF NEW CASTLE

The town of New Castle developed its water-supply system via a connection to the New York City Catskill Aqueduct. Suppling itself and the town of Mt. Pleasant.


ASHOKAN RESERVOIR

**1980** — GROWING PAINS


CROTON AQUEDUCT

New Castle is met with a growing-need for a more robust water supply. Giving way for a new pipe relining project to commence, resulting in a new pump station and water treatment plant, as well as using Croton Water Supply System's as a secondary-source of water.

**1993** — THE MILLWOOD TREATMENT PLANT

The treatment plant was put on line, pumping untreated water from the Catskill & Croton water-supply systems, to then be redistributed throughout New Castle.


WATER TREATMENT PLANT

SOURCES:
HTTPS://DIGITALCOLLECTIONS.NYPL.ORG/COLLECTIONS/PHOTOGRAPHS-OF-THE-CATSKILL-WATER-SUPPLY-SYSTEM-IN-PROCESS-OF-CONSTRUCTION#/?TAB=ABOUT&SCROLL=20
HTTPS://WWW.NCBI.NLM.NIH.GOV/BOOKS/NBK566285/

The New York State Department of Environmental Conservation has called Pace Pleasantville drinking water the "'champagne' of drinking water" that "consistently wins annual taste tests"? Well, almost. In fact, our water mainly originates from pristine reservoirs in the Catskill Mountains and the Delaware River watershed that carry ancient Indian names such as Ashokan and Pepacton.

Thank New York City. By law, Pace Pleasantville is entitled to share in the water the city distributes "from 19 reservoirs and three controlled lakes spread across a nearly 2,000-square-mile watershed. . . located upstate in portions of the Hudson Valley and Catskill Mountains that are as far as 125 miles north of the city," as the city describes in its Annual Water Quality Report.

By law, any village, town or city through which the New York City aqueducts travel is entitled to tap into the water supply. Therefore, nearby New Castle does so and distributes it to the Town of Mt. Pleasant for which our Pleasantville campus is a customer.

New York City's water supply (our water supply) is an engineering marvel, as the New York City Department of Environmental Protection describes. "The watersheds of the three systems cover an area of almost 2,000 square miles, approximately the size of the State of Delaware. The reservoirs combined have a storage capacity of 550 billion gallons. The water flows to New York City through aqueducts, and 97 percent reaches homes and businesses through gravity alone." The massive aqueducts run mostly beneath ground on the west side of the Hudson River and then beneath the river bottom to the east side of the river and then south to us.

In the earliest days of our water supply history, the city's lone reservoir was in the middle of Manhattan. When contamination due to 19th century life of poor sanitation and garbage disposal, including waste from horses and other animals, led to cholera outbreaks, the city turned its sights north to the Croton River. Then, as population grew, it turned to the Catskills and Delaware.

Before that water reaches our campus taps, Pace University, Town of New Castle, and New York City together perform hundreds of thousands of analyses for tens of thousands of samples every year. As we write in this year's Annual Water Quality Report, Pace Pleasantville drinking water "met all State drinking water health standards. We are proud to report that our system did not violate a maximum contaminant level or any other water quality standard."

Pace Pleasantville water originates in the Catskill Mountains, from New York City's Ashokan Reservoir, 91 miles away in Ulster County, NY. The Ashokan's water flows down the west side of the Hudson River through deep underground aqueducts, then underneath the river and down the east side to the Town of Mount Pleasant and then the Town of New Castle Water District and to our campus.

Our water source is purchased from the Town of Mount Pleasant which purchases its water from the Town of New Castle Water District. New Castle's primary source is the Catskill Aqueduct System, and its secondary source is the Croton Aqueduct System. During 2024, our system did not experience any restriction of our water source.

New York City's Croton Reservoir, 12 miles from campus in Westchester County, is our secondary or supplemental source, used in the event of emergencies, shortages, or maintenance activities in the Catskill aqueduct.

It was created by damming the Croton River. Its earliest configuration, Croton Lake, created in 1842, was the city's first upstate water source.

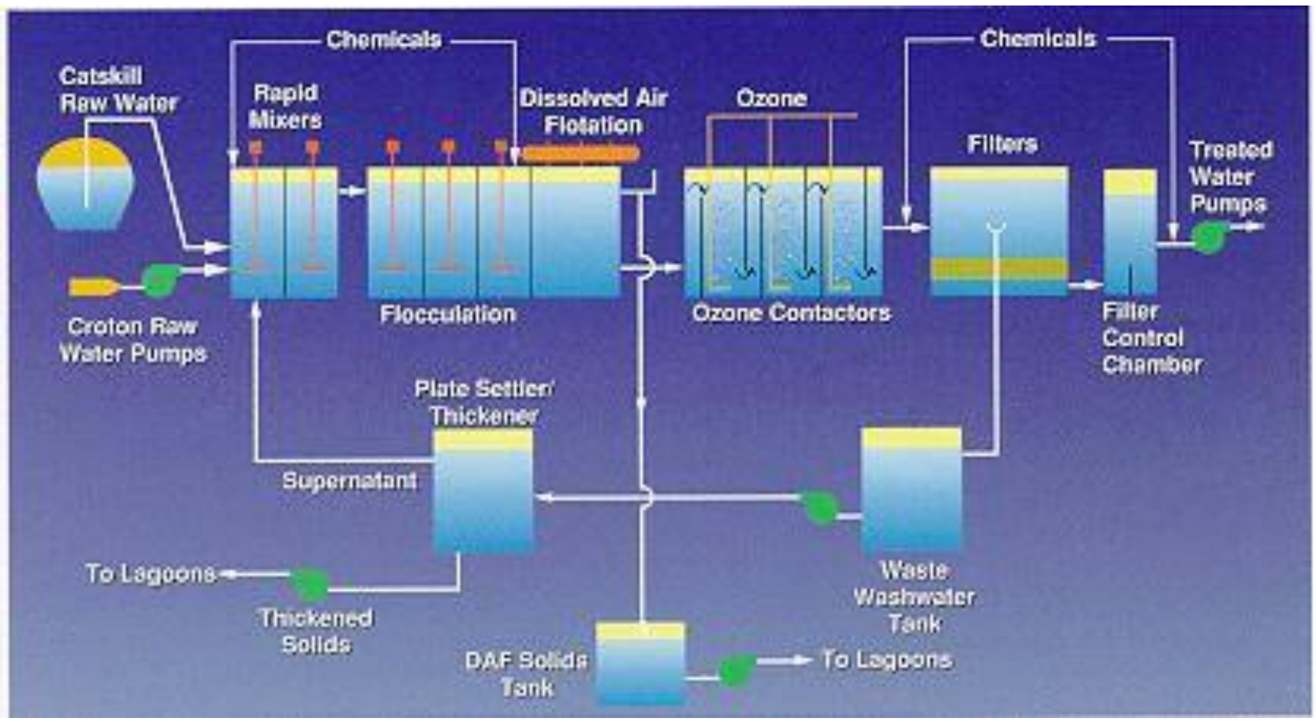
In general, the sources of drinking water (both tap water and bottled water) include rivers, lakes, streams, ponds, reservoirs, springs, and wells. As water travels over the surface of the land or through the ground, it dissolves naturally occurring minerals and, in some cases, radioactive material, and can pick up substances resulting from the presence of animals or from human activities.

Because Pace Pleasantville manages the local collection and distribution of its drinking water, we are considered a “community water system” under federal and state law. Pace must therefore comply with applicable governmental testing and reporting requirements.

In order to ensure that tap water is safe to drink, the State and the EPA prescribe regulations which limit the amount of certain contaminants in water provided by public water systems. The State Health Departments and the FDA’s regulations establish limits for contaminants in bottled water which must provide the same protection for public health.

**TREATING OUR DRINKING WATER**

Before water arrives at Pace, it is treated in New Castle (https://mynewcastle.org/213/Water-Treatment). New Castle performs treatment to maintain compliance with the Surface Water Treatment Rule (SWTR) under the authority of the New York State Health Department and the U.S. Environmental Protection Agency. New Castle performs the following treatment:




Filtration

The Town of New Castle processes the raw water received from New York City through its Millwood Water Treatment Plant. The treatment process includes flocculation, dissolved air flotation, ozone disinfection and final filtration (see below for more information).

Chlorination

Chlorine is added for final disinfection and to provide a detectable level of chlorine residual throughout the distribution system. The chlorine residual is required in order to maintain bacteria free water.

## Corrosion Control Treatment

In order to inhibit leaching of lead and copper from household plumbing, New Castle adds caustic soda for pH adjustment and polyphosphate to inhibit corrosion, after filtration.

## Fluoridation

Fluoride is added to the finished water for prevention of dental cavities.

## Rapid Mixers

Chemicals such as alum, chlorine, caustic soda, and more are added to the water and mixed for about one minute to help with cleaning.

## Flocculation

Water is mixed for 30 minutes to form “floc” particles (clumps of dirt, algae, bacteria, etc.) that are easier to remove.

## Dissolved Air Flotation (DAF)

Microscopic air bubbles float the floc particles to the surface, where they are skimmed off. This process takes about 25 minutes.

## Ozone Treatment

Ozone, a strong disinfectant, is added to kill bacteria, viruses, and other microorganisms. The process takes six minutes.

## Filtration

Water passes through filters (anthracite coal and sand) to remove remaining particles and floc.

## Final Chemical Treatment

A small amount of chlorine is added to keep the water clean during distribution. Caustic soda raises the pH, making the water slightly basic. Corrosion inhibitors are added to protect pipes, and fluoride is included for dental health.

## Quality Control

Water quality is continuously monitored through computerized systems, regular testing, and backup systems for emergencies.

The water is treated through a five-stage process that typically removes 98.14% of particulate matter. Anything left behind is disinfected through the ozonation process prior to distribution.

Treatment or other requirements that a water system must follow is triggered in the event that the concentration of a contaminant exceeds its Action Level (AL)

## FACTS AND FIGURES

Our water system serves 2,753 consumers with 34 service connections at the Pace University Pleasantville, New York campus. The total water withdrawn by the New Castle Water District in 2024 was 1,176.150 million gallons (MG) of raw water from the Catskill Aqueduct System and 1.029 MG from the Croton Aqueduct system. 367.255 MG were supplied to the Village of Pleasantville, from where Pace University purchases its water. The total amount of water that was delivered to Pace University from Pleasantville in 2024 was 18.128 million gallons

## TESTING FOR DRINKING WATER QUALITY

### SAMPLING AND MONITORING

Your drinking water is sampled at the New Castle Water District. The water is sampled by having ten samples collected from our water system. On a frequency basis, the State allows us to test for some contaminants less than once per year because the concentrations of these contaminants do not change frequently. Some of our data, though representative, are more than one year old. Field turbidity readings are recorded 5 days a week. The samples for these in-house testing are taken from varying locations on-site. Testing for total coliform is done 3 times a month by the State. Throughout the month, the three testing dates fall between the 1 – 7th, 8 – 14th, and 15 – 21st of the month. Disinfection byproducts are tested 4 times a month, with samples taken at two different sites. Lead and copper may be tested every 6 months. However, if no contamination is found, the testing frequency would change to annual testing or testing every 3 years if there continues to be no contamination. Since Pace is a recipient of water that is already treated, we are not required to monitor additional contaminants.

**More information on the parameters being monitored can be found in the Glossary at the end of this report.**

Pace University is required to monitor for:

- Total Coliform
- Lead
- Copper
- Total Trihalomethanes
- Haloacetic Acids
- Turbidity

Additional contaminants are monitored in the New Castle Water District, including microbiological contaminants, inorganic compounds, volatile organic compounds, radiological compounds, and synthetic organic compounds.

Contaminants that may be present in source water include microbial contaminants; inorganic contaminants; pesticides and herbicides; organic chemical contaminants; and radioactive contaminants.

REGULATION OF DRINKING WATER

Regulations under the New York State Sanitary Code and the Federal Safe Drinking Water Act require Pace University to routinely test your drinking water for numerous contaminants.

Federal Safe Drinking Water Act - Sets standards for drinking water quality, including the Minimum Contaminant Level and reporting.

Environmental Protection Agency (EPA) - When Congress writes an environmental law, the EPA develops and enforces regulations. [https://www.epa.gov/]

Consumer Confidence Reports - All community water systems must prepare and distribute annual reports about the water they provide, including information on detected contaminants, possible health effects, and the water's source.

Freedom of Information Act (FOIA) - Provides the public the right to request access to records from any federal agency. Federal agencies are required to disclose any information requested under the FOIA unless it falls under one of nine exemptions which protect interests such as personal privacy, national security, and law enforcement. [https://www.foia.gov/]

<u>The Public Notification Rule</u> (PN) - Ensures that consumers will know if there is a problem with their drinking water. Alerts consumers:

- if there is risk to public health
- if the water does not meet drinking water standards
- if the water system fails to test its water
- if the system has been granted a variance (use of less costly technology)
- if the system has been granted an exemption (more time to comply with a new regulation)

Along with defining and setting requirements, The New York State Sanitary Code and EPA also established a 3-tier notification system for emergency reporting. The 3 Tiers of Public Notification are described in the table below. More definitions can be found in the glossary at the end of this report.

| Tier | When? | Distribution Time | Delivery Method |
|---|---|---|---|
| 1 | Human health may be immediately impacted | 24 hours | media, posting, or through the mail |
| 2 | Contaminant exceedance but no immediate risk to human health | 30 days | media, posting, or through the mail |
| 3 | Violation does not have a direct impact on human health (EX: failing to take a required sample on time) | 1 year | (Required) media outlets (television, radio, newspapers), post notice in public places, personally deliver a notice to their customers, or an alternative method |

## CONSERVING OUR SUPPLY

What are regular precautions I should take?

If the faucet or showerhead has not been used recently, defined as a week or more, the CDC recommends flushing them before use. It is recommended that you run the cold water for two minutes, then switch to hot water until the water starts to feel hot. Regularly clean any device that uses water. Read more on the CDC's website.

There are steps that you as a Pace Community member can do to protect our water source. Don't dispose of hazardous waste (i.e. mothballs, cleaners, medication) down your sink, on the ground, or in storm sewers. Help clean up local water sources.

DOs & DON'Ts of Water Conservation

In or out of a drought, hundreds of gallons of water can be saved each week by following these simple water-saving tips.

BATHROOM

- Do take short showers and save 5 to 7 gallons a minute.
- Do fill the tub halfway and save 10 to 15 gallons.
- Do install water-saving toilets, shower heads and faucet aerators. Place a plastic bottle filled with water in your toilet tank if you can't switch to a low flow toilet.
- Don't run the water while shaving, washing your hands or brushing your teeth. Faucets use 2 to 3 gallons a minute.
- Don't use the toilet as a wastebasket, and don't flush it unnecessarily.

OUTDOORS

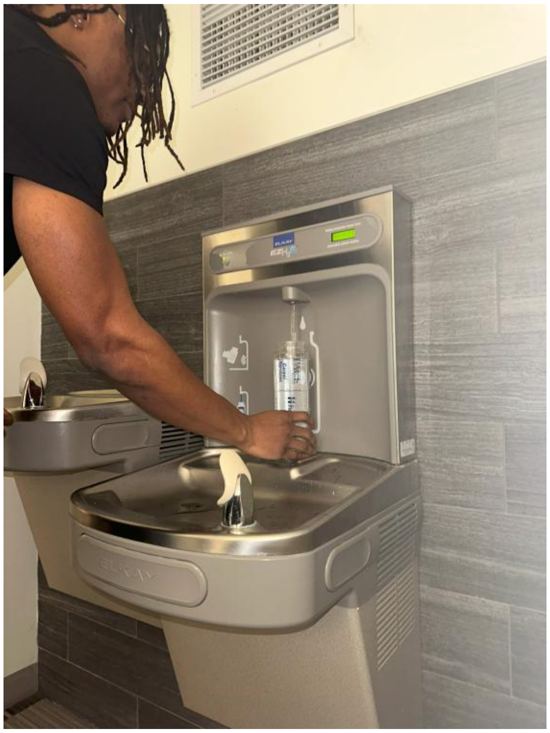
- Do use a self-closing nozzle on your hose.
- Don't water your sidewalk or driveway-sweep them clean.
- Don't over water your lawn or plants. Water before 9 a.m. or after 7 p.m.

KITCHEN & LAUNDRY

- Do run the dishwasher and washing machine only when full. Save even more by using the short cycle.
- Do install faucet aerators.
- Don't let the water run while washing dishes. Kitchen faucets use 2 to 3 gallons a minute. Filling a basin only takes 10 gallons to wash and rinse.
- Don't run water to make it cold. Have it chilled in the refrigerator, ready to drink.

EVERYWHERE

- Do repair leaky faucets and turn taps off tightly. A slow drip wastes 15 to 20 gallons each day.
- Don't open fire hydrants.



## INFORMATION ON FLUORIDE ADDITION

Our system is one of the many drinking water systems in New York State that provides drinking water with a controlled, low level of fluoride for consumer dental health protection. Fluoride is added to your water by the New Castle/Stanwood Consolidated Water District. According to the United States Centers for Disease Control, fluoride is very effective in preventing cavities when present in drinking water at a properly controlled level. To ensure that the fluoride supplement in your water provides optimal dental protection, New Castle/Stanwood Consolidated Water District, Pace University's water supplier, monitors fluoride levels on a daily basis to make sure fluoride is maintained at a target level of 0.7 mg/L. During 2024, monitoring showed fluoride levels in your water were within 0.1 mg/L of the target level 91.34 % of the time. None of the monitoring results showed fluoride at levels that approach the 2.2 mg/l MCL for fluoride.

## INFORMATION ON CRYPTOSPORIDIUM AND GIARDIA

Cryptosporidium and Giardia are microbial pathogens found in surface water and groundwater under the influence of surface water. During 2024, we were not required to monitor for these organisms. However, our supplier (NYCDEP) found very low, sporadic levels of Crypto and Giardia. Therefore, the testing indicates a possible small presence of these organisms in our raw, untreated water. Furthermore, our water passes through processes at the Millwood Water Treatment Plant and is very aggressively treated.

Cryptosporidium and Giardia must be ingested to cause disease, and it may spread through other means other than drinking water. Ingestion of Cryptosporidium may cause cryptosporidiosis, a gastrointestinal infection. Symptoms of infection include nausea, diarrhea, and abdominal cramps. Most healthy individuals can overcome disease within a few weeks. However, immuno-compromised people are at greater risk of developing life-threatening illness. Ingestion of Giardia may cause giardiasis, an intestinal illness. People exposed to Giardia may experience mild or severe diarrhea, or in some instances no symptoms at all. Fever is rarely present. Occasionally, some individuals will have chronic diarrhea over several weeks or a month, with significant weight loss. Giardiasis can be treated with anti-parasitic medication. Individuals with weakened immune systems should consult with their health care providers about what steps would best reduce their risks of becoming infected with Giardiasis. Individuals who think that they may have been exposed to Giardiasis should contact their health care providers immediately. The Giardia parasite is passed in the feces of an infected person or animal and may contaminate water or food. Person to person transmission may also occur in day care centers or other settings where hand-washing practices are poor.

Ozone is one of the most effective disinfectants for Cryptosporidium and Giardia, and New Castle water is both ozonated and filtered to minimize any health risk from these organisms. For additional information on Cryptosporidiosis or Giardiasis, please call the Westchester County Department of Health at (914) 813-5000 or write the Westchester County Department of Health, 25 Moore Avenue, Mount Kisco, New York 10549.

## HOW CAN I LIMIT MY LEAD EXPOSURE?

**RUN YOUR TAP**
for 30 seconds to 2 minutes before using water for drinking or cooking, when your water has been sitting for several hours.

**Use Cold Water**
for cooking, drinking, or preparing infant formula. Hot tap water is more likely to contain lead and other metals.

**Remove & Clean**
the faucet screen monthly (also called an aerator), where small particles can get trapped.

**Hire**
a licensed plumber to identify and replace plumbing fixtures and/or service line that contain lead.

## IS THERE LEAD IN MY DRINKING WATER?

Lead can cause serious health problems, especially for pregnant women and young children. Lead in drinking water is primarily from materials and components associated with service lines and home plumbing. The Pace University - Pleasantville water supply is responsible for providing high quality drinking water and removing lead pipes, but cannot control the variety of materials used in plumbing components in your home.

You share the responsibility for protecting yourself and your family from the lead in your home plumbing. You can take responsibility by identifying and removing lead materials within your home plumbing and taking steps to reduce your family's risk. Before drinking tap water, flush your pipes for several minutes by running your tap, taking a shower, doing laundry or a load of dishes. You can also use a filter certified by an American National Standards Institute accredited certifier to reduce lead in drinking water. If you are concerned about lead in your water and wish to have your water tested, contact the Pace University - Pleasantville water supply. Information on lead in drinking water, testing methods, and steps you can take to minimize exposure is available at http://www.epa.gov/safewater/lead.

## IS OUR WATER SYSTEM MEETING OTHER RULES THAT GOVERN OPERATIONS?

We are required to monitor your drinking water for specific contaminants on a regular basis. Results of regular monitoring are an indicator of whether or not your drinking water meets health standards. During 2024, our system was in compliance with applicable State drinking water operating and reporting requirements.

## ARE THERE CONTAMINANTS IN OUR DRINKING WATER?

As the State regulations require, we routinely test your drinking water for numerous contaminants. Pace University is required to monitor for total coliform, lead, copper, total trihalomethanes, haloacetic acids and turbidity. Additional contaminants are monitored in the New Castle Water District, including microbiological contaminants, inorganic compounds, volatile organic compounds, radiological compounds, and synthetic organic compounds. The table presented below depicts which compounds were detected in your drinking water. The State allows us to test for some contaminants less than once per year because the concentrations of these contaminants do not change frequently. Some of our data, though representative, are more than one year old.

It should be noted that all drinking water, including bottled drinking water, may be reasonably expected to contain at least small amounts of some contaminants. The presence of contaminants does not necessarily indicate that water poses a health risk. More information about contaminants and potential health effects can be obtained by calling the EPA's Safe Drinking Water Hotline (800-426-4791) or the Westchester County Health Department at (914) 813-5000.

**TABLE OF DETECTED CONTAMINANTS**

| Contaminant | Violation? Yes/No | Date of Sample | Level Detected (Avg/Max) (Range) | Unit Measurement | MCLG | Regulatory Limit (MCL, TT, or AL) | Likely Source of Contamination |
|---|---|---|---|---|---|---|---|
| **Inorganic Contaminants** | | | | | | | |
| Alkalinity | No | 6/2024 | 12.8 | mg/l | N/A | N/A | Erosion of natural deposits. |
| Barium | No | 6/2024 | 0.0079 | mg/l | 2.0 | 2.0 | Erosion of natural deposits. |
| Calcium | No | 6/2024 | 4.76 | mg/l | N/A | N/A | Erosion of natural deposits. |
| Chloride | No | 5/2024 | 9.9 | mg/l | N/A | 250 | Naturally occurring or indicative of road salt contamination. |
| Corrosivity by Calculation | No | 2019 | -2.19 | mg/l | N/A | N/A | Erosion of natural deposits. |
| Fluoride | No | 2024 | 0.89 [1] (0.44-0.89) | mg/l | N/A | 2.2 | Erosion of natural deposits; water additive that promotes strong teeth. |

**TABLE OF DETECTED CONTAMINANTS**

| Contaminant | Violation? Yes/No | Date of Sample | Level Detected (Avg/Max) (Range) | Unit Measurement | MCLG | Regulatory Limit (MCL, TT, or AL) | Likely Source of Contamination |
|---|---|---|---|---|---|---|---|
| Hardness [2] | No | 6/2024 | 11.9 | mg/L | N/A | N/A | Naturally occurring. |
| Nickel | No | 6/2024 | 0.33 | ug/l | N/A | N/A | Naturally occurring. |
| pH | No | 6/2024 | 7.49 | units | N/A | N/A | N/A. |
| Sodium | No | 6/2024 | 9.15 | mg/l | N/A | See health effect [3] | Naturally occurring; Road salt; Animal waste. |
| Sulfate | No | 6/2024 | 2.82 | mg/l | N/A | 250 | Naturally occurring. |
| Total Dissolved Solids | No | 12/2023 | 55.6 | mg/l | N/A | N/A | Erosion of natural deposits. |
| Zinc | No | 6/2024 | 0.0047 | mg/l | N/A | 5 | Naturally occurring; Mining waste. |
| **Radioactive Contaminants** | | | | | | | |
| Gross Alpha (Including radium-226 but excluding radon and uranium) | No | 4/2018 | 0.464 | pCi/L | 0 | 15 | Erosion of natural deposits. |
| Beta particles and photon activity from man-made radionuclides | No | 4/2018 | 0.923 | pCi/L | 0 | 50 [4] | Decay of natural deposits and man-made emissions. |
| Combined radium-226 and 228 | No | 4/2018 | 1.086 | pCi/L | 0 | 5 | Erosion of natural deposits. |
| Uranium | No | 4/2018 | 0.125 | ug/L | 0 | 30 | Erosion of natural deposits. |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Disinfection Byproducts** [5] | | | | | | | |
| Total Haloacetic Acids | No | 2024 | 16.28 [6] (11.1 – 19.5) [7] | ug/L | N/A | 60 | By-product of drinking water disinfection needed to kill harmful organisms. [8] |
| Total Trihalomethanes | No | 2024 | 41.97[6] (21.15 – 58.73) [7] | ug/L | N/A | 80 | By-product of drinking water chlorination needed to kill harmful organisms. TTHMs are formed when source water contains large amounts of organic matter. [8] |
| **Disinfectants** | | | | | | | |
| Chlorine Residual | No | 2024 | 1.3 [9] (0.8 -1.7) | mg/L | N/A | 4 [10] | Water additive used to control microbes. |
| **Microbiological Contaminants** | | | | | | | |
| Distribution Turbidity [11] | No | 2024 | 0.26 (0.05 - 1.3) [12] | NTU N/A >5 | Soil | runoff. | |
| **Synthetic Organic Contaminants** [13] | | | | | | | |
| Perfluorooctanoic Acid (PFOA) | No | 10/2024 5/2024 | 3.30 (ND – 3.30) | ng/L | N/A | 10 | Released into the environment from widespread use in commercial and industrial applications |
| Perfluorooctanesulfonic Acid (PFOS) | No | 10/2024 5/2024 | 2.80 (ND – 2.80) | ng/L | N/A | 10 | Released into the environment from widespread use in commercial and industrial applications |

## TABLE OF UNREGULATED DETECTED SUBSTANCES

| Contaminant | Violation Yes/No | Date of Sample | Level Detected (Avg/Max) (Range) | Unit Measurement | MCLG or Health Advisory Level[14] | Regulatory Limit (MCL, TT, or AL) | Likely Source of Contamination |
|---|---|---|---|---|---|---|---|
| **Disinfection Byproducts** | | | | | | | |
| Bromochloroacetic Acid | No | 2024 | 1.4 (ND – 1.4) | ug/L | N/A | N/A | By-product of drinking water chlorination needed to kill harmful organisms. |

Notes:

1. This level represents the highest of the detected fluoride levels and range of values detected in 2024.
2. Soft water: 0-45 mg/L. Soft to moderately hard water: 46-90 mg/L. Moderately hard to hard water: 91-130 mg/L.
3. Water containing >20mg/L of sodium should not be used for drinking by people on severely restricted sodium diet. Water containing >270 mg/L of sodium should not be used for drinking by people on moderately restricted sodium diet.
4. The State considers 50 pCi/L to be the level of concern for Beta particles.
5. Disinfection byproducts data is collected from two different sampling points from 1st quarter 2024 to 4th quarter 2024.
6. The level represents the highest value of locational running annual average calculated from the data collected.
7. Range represents the minimum and maximum values out of all TTHMs and HAA5s collected throughout the year.
8. An additional source of these TTHMs and Haloacetic acids is from the chlorination of water provided by the Mount Pleasant Water District. (TTHMs include chloroform, bromodichloromethane, dibromochloromethane, and bromoform; Haloacetic acids include mono-, di-, and tri-chloroacetic acid, and mono- and di-bromoacetic acid).
9. The value presented is the annual average for 2024, as well as the range of values.
10. The value presented represents the Maximum Residual Disinfectant Level (MRDL) which is a level of disinfectant added for water treatment that may not be exceeded at the consumer's tap without an unacceptable possibility of adverse health effects.
11. Turbidity is a measure of the cloudiness of the water found in the distribution system. We monitor it because it is a good indicator of water quality. High turbidity can hinder the effectiveness of disinfectants. Our highest single turbidity measurement for the year occurred in November 2024 (1.3 NTU). State regulations require that turbidity must always be less than or equal to 5.0 NTU. The regulations require that 95% of the turbidity samples collected have measurements below 0.5 NTU. Although November was the month when we had the fewest measurements meeting the treatment technique for turbidity, the levels recorded were within the acceptable range allowed and did not constitute a treatment technique violation.
12. The level presented represents the yearly turbidity average and the range of detected turbidity levels.

13. PFOA and PFOS are part of a larger group of chemicals referred to as perfluoroalkyl substances (PFASs). PFAS are manmade chemicals that have been widely used in various consumer, commercial, and industrial products since the 1950s. These chemicals' unique properties make them resistant to heat, oil, stains, grease, and water and useful in a wide variety of everyday products. One of the PFAS' was widely used in fire-fighting foam. On August 26, 2020, New York State adopted new drinking water standards for public water systems that set maximum contaminant levels (MCLs) of 10 parts per trillion (10 ppt) each for perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS), and 1 part per billion (1 ppb) for 1,4-dioxane.

    PFOA and PFOS have caused a wide range of health effects when studied in animals that were exposed to high levels. The most consistent findings in animals were effects on the liver and immune system and impaired fetal growth and development. PFOA and PFOS also cause cancer in laboratory animals exposed to high levels over their lifetimes. Additional studies of exposures PFOA and PFOS in people provide evidence that some of the health effects seen in animals may also occur in humans. The levels presented in the table represent the highest detected PFAS level, and the range of values in 2024.
14. USEPA Health Advisory Levels identify the concentration of a contaminant in drinking water at which adverse health effects and/or aesthetic effects are not anticipated to occur over specific exposure durations. Health Advisory Levels are not to be construed as legally enforceable federal standards and are subject to change as new information becomes available.

## LEAD AND COPPER

| Contaminant | Violation (Yes/No) | Date of Sample | Level Detected (90th% value range) | Unit Measurement | Regulatory Limit (AL) | MCLG | # of samples collected | # of samples exceeds AL Range | Likely Source of contamination |
|---|---|---|---|---|---|---|---|---|---|
| Lead [1] | No | 2024 | ND (ND -1.4) | ug/L | 15 | 0 | 10 | 0 | Corrosion of household plumbing systems; Erosion of natural deposits. |
| Copper [2] | No | 2024 | 0.033 (0.005 -0.048) | mg/L | 1.3 | 1.3 | 10 | 0 | Corrosion of household plumbing systems; Erosion of natural deposits. |

Notes:

1. The level presented represents the 90$^{th}$ percentile of the 10 sites tested. A percentile is a value on a scale of 100 that indicates the percent of a distribution that is equal to or below it. The 90$^{th}$ percentile is equal to or greater than 90% of the lead values detected at your water system. In this case, 10 samples were collected at your water system and the 90$^{th}$ percentile value was the ninth highest value. The action level for lead was not exceeded at any of the sites tested in 2024.
2. The level presented represents the 90$^{th}$ percentile of the 10 samples collected. The action level for copper was not exceeded for the 10 sites tested in 2024.

## WHAT DOES THIS INFORMATION MEAN?

As you can see by the table, our system had no violations. We have learned through our testing that some contaminants have been detected; however, these contaminants were detected below New York State requirements.

## Lead

It should be noted that the action level for lead was not exceeded in any of the 10 samples collected. However, we are required to present the following information on lead in drinking water:

Lead can cause serious health effects in people of all ages, especially pregnant people, infants (both formula-fed and breastfed), and young children. Lead in drinking water is primarily from materials and parts used in service lines and in home plumbing. The Pace University Pleasantville water supply is responsible for providing high quality drinking water and removing lead pipes but cannot control the variety of materials used in the plumbing in your home. Because lead levels may vary over time, lead exposure is possible even when your tap sampling results do not detect lead at one point in time. You can help protect yourself and your family by identifying and removing lead materials within your home plumbing and taking steps to reduce your family's risk. Using a filter, certified by an American National Standards Institute accredited certifier to reduce lead, is effective in reducing lead exposures. Follow the instructions provided with the filter to ensure the filter is used properly. Use only cold water for drinking, cooking, and making baby formula. Boiling water does not remove lead from water. Before using tap water for drinking, cooking, or making baby formula, flush your pipes for several minutes. You can do this by running your tap, taking a shower, doing laundry or a load of dishes. If you have a lead service line or galvanized requiring replacement service line, you may need to flush your pipes for a longer period. If you are concerned about lead in your water and wish to have your water tested, contact the Pace University Facilities Management Office at (914) 923-2840. Information on lead in drinking water, testing methods, and steps you can take to minimize exposure is available.

## FAQ

*Who is responsible for maintenance of the water supply?*

The water & wastewater specialists of Allied Pollution Control Inc. are Pace University's water system operators. Pace's water source is purchased from the Town of Mount Pleasant which purchases its water from the Town of New Castle Water District, whose primary source is the Catskill Aqueduct System and its secondary is the Croton Aqueduct System.

*How and where is water sampled? What frequency?*

The water is sampled at the New Castle Water District. The water is sampled by having ten samples collected from our water system. On a frequency basis, the State allows us to test for some contaminants less than once per year because the concentrations of these contaminants do not change frequently. Some of our data, though representative, are more than one year old. Field turbidity readings are recorded 5 days a week. The samples for these in-house testing are taken from varying locations on-site. Testing for total coliform is done 3 times a month by the State. Throughout the month, the three tasting dates fall between the 1 – 7th, 8 – 14th, and 15 – 21st of the month. Disinfection byproducts are tested 4 times a month, with samples taken at two different sites. Lead and copper may be tested every 6 months. However, if no contamination is found, the testing frequency would change to annual testing or testing every 3 years if there continues to be no contamination. Since Pace is a recipient of water that is already treated, we are not required to monitor additional contaminants.

*What is tested?*

Pace University is required to monitor for total coliform, lead, copper, total trihalomethanes, haloacetic acids, and turbidity. Additional contaminants are monitored in the New Castle Water District, including microbiological contaminants, inorganic compounds, volatile organic compounds, radiological compounds, and synthetic organic compounds.

*What are the regulations Pace has to follow?*

Regulations under the New York State Sanitary Code and the Federal Safe Drinking Water Act require Pace University to routinely test your drinking water for numerous contaminants.

*What is Pace's Water Emergency Management & Response Plan?*

Pace University can send notifications through Pace Alert, the university's emergency notification system (via e-mail, text, and Pace Safe App). Students, staff, and faculty are automatically enrolled. To update your preferences, see Emergency Notifications.

*How can I stay informed about water quality emergencies?*

Annual Water Quality Reports are published by Facilities and Capital Projects every May. Reports are published on Facility Updates page. In the future, this page will display updated water quality data so you can make informed decisions.

*What should I do in the event of a water emergency? Where can I report concerns?*

In the event of a water emergency, if you have any questions concerning your drinking water, please contact the Pace University Facilities Management Office at (914) 923-2840. We want you to be informed about your drinking water. To obtain additional information on the drinking water provided to Pace University, you may contact the Mount Pleasant Water District directly, at 119 Lozza Drive, Valhalla, New York 10595-1268, or by telephone at (914) 742-2313.

*What are regular precautions I should take?*

If the faucet or showerhead has not been used recently, defined as a week or more, the CDC recommends flushing them before use. It is recommended that you run the cold water for two minutes; then switch to hot water until the water starts to feel hot (CDC). Regularly clean any device that uses water (CDC). Read more on the CDC's website. There are steps that you as a Pace Community member can do to protect our water source. Don't dispose of hazardous waste (i.e. mothballs, cleaners, medication) down your sink, on the ground, or in storm sewers. Help clean up local water sources (EPA).

*What are my rights?*

Since 2010 the UN has declared that clean water is a human right. Furthermore, one needs to know what's in their water to make informed decisions, a human right recognized by the UN in 2015. You can read more about this on the UN's website. Current federal regulations (§ 141.151) only require reports to be posted yearly – having water quality that could be a year old can not be used to make informed decisions. Blue CoLab seeks to give regular reports so community members can actively make informed decisions on their water.

*Where does Pace wastewater go?*

Wastewater from most of the Pocantico River watershed goes to the Yonkers plant, which is the largest contributor of liquid waste/sewage to the Lower Hudson and is also a combined sewer system.

## GLOSSARY

**Maximum Contaminant Level (MCL):** The highest level of a contaminant that is allowed in drinking water. MCLs are set as close to the MCLGs as feasible.

**Maximum Contaminant Level Goal (MCLG):** The level of a contaminant in drinking water below which there is no known or expected risk to health. MCLGs allow for a margin of safety.

**Maximum Residual Disinfectant Level (MRDL):** The highest level of a disinfectant allowed in drinking water. There is convincing evidence that addition of a disinfectant is necessary for control of microbial contaminants.

**Maximum Residual Disinfectant Level Goal (MRDLG):** The level of a drinking water disinfectant below which there is no known or expected risk to health. MRDLGs do not reflect the benefits of the use of disinfectants to control microbial contamination.

**Action Level (AL):** The concentration of a contaminant which, if exceeded, triggers treatment or other requirements which a water system must follow.

**Treatment Technique (TT):** A required process intended to reduce the level of a contaminant in drinking water.

**Non-Detects (ND):** Laboratory analysis indicates that the constituent is not present.

**Nephelometric Turbidity Unit (NTU):** A measure of the clarity of water. Turbidity in excess of 5 NTU is just noticeable to the average person.

**Milligrams per liter (mg/l):** Corresponds to one part of liquid in one million parts of liquid (parts per million - ppm).

**Micrograms per liter (ug/l):** Corresponds to one part of liquid in one billion parts of liquid (parts per billion - ppb).

**Nanograms per liter (ng/l):** Corresponds to one part of liquid to one trillion parts of liquid (parts per trillion - ppt).

**Picograms per liter (pg/l):** Corresponds to one part per of liquid to one quadrillion parts of liquid (parts per quadrillion – ppq).

**Picocuries per liter (pCi/L):** A measure of the radioactivity in water.

**Millirems per year (mrem/yr):** A measure of radiation absorbed by the body.

**Million Fibers per Liter (MFL):** A measure of the presence of asbestos fibers that are longer than 10 micrometers.

(from https://bluecolab.pace.edu/pacewater/)

## Total Coliform

Total coliforms are a group of bacteria that includes non-fecal coliform (found in soil and surface water) and fecal coliform. Fecal coliform is a subset of coliform that is heat tolerant and found in animal waste and human sewage. This group consists of Escherichia coli (E. coli), which is a bacteria that is commonly found in the intestines of animals and humans.

How it's Tested: A method of determining bacteria contamination in water is to count the number of bacteria colonies that grow on a prepared medium. The number of total coliform bacteria is widely used as an indicator for drinkable water in the United States. E. coli present in water is a strong indicator of sewage or animal waste contamination. Total coliforms were originally believed to indicate the presence of fecal contamination, but they have been found to be widely distributed in nature, not always associating with the gastrointestinal tract of warm-blooded animals. However, high numbers of even harmless bacteria can indicate high numbers of harmful bacteria, viruses, or protozoa[1], as well.

Effect on Health: Sewage and animal waste can contain organisms that cause disease, which may result in severe illness if consumed. While minor gastrointestinal discomfort is one of the most common symptoms, pathogens that cause only minor sickness in some people may cause serious conditions or death in others, especially in the very young, old, or those with weakened immune systems. Read more from The United States Geological Survey (USGS).

## Lead

The likely source of lead contamination in water is from corrosion of household plumbing systems and erosion of natural deposits. It is possible that levels of lead at different homes in the community may be higher than others as a result of materials used in the home's plumbing. Pace University's water system is responsible for providing high-quality drinking water but cannot control the variety of materials used in plumbing components. To minimize the potential for lead exposure, you can flush your tap for 30 seconds to 2 minutes before using water for drinking or cooking.

How it's Tested: The Safe Drinking Water Act requires the U.S. Environmental Protection Agency (EPA) to set up goals for the level of contaminants in drinking water below which there is no known or expected risk to health with an adequate margin of safety. These goals are solely based on possible health risks and are called maximum contaminant level goals (MCLGs)[2]. EPA has set the MCLG for lead in drinking water at zero because even at low exposure levels, lead is a toxic metal that can be harmful to human health.

Effect on Health: Lead is persistent and can accumulate in the body over time. Elevated levels of lead can cause serious health problems, especially for pregnant women, infants, and young children. In children, low levels of exposure have been linked to damage to the central and peripheral nervous system, learning disabilities, slowed growth, impaired hearing, and impaired formation and function of blood cells such as anemia.

Read more from the US Environmental Protection Agency (EPA).

## Copper

Like lead, copper primarily enters drinking water through the corrosion of plumbing materials and erosion of natural deposits. Additionally, copper enters the environment from industrial and domestic waste, mining, and mineral leaching.

How it's Tested: Similar to lead, the U.S. Environmental Protection Agency (EPA) must determine a maximum contaminant level goal (MCLG)[2] for copper in water. If the concentration of copper exceeds an action level of 1.3 parts per million (ppm) in more than 10% of the taps sampled, a number of additional actions must be undertaken to control corrosion. Action level (AL)[3] is the concentration of a contaminant that, if exceeded, triggers treatment or other requirements that a water system must follow.

Effect on Health: Exposure to copper may cause stomach and intestinal distress, liver and kidney damage, and in high doses, anemia and brain damage.

Read more from the US Environmental Protection Agency (EPA) and The United States Geological Survey (USGS).

### Haloacetic Acids

Similar to trihalomethanes, haloacetic acids (HAA5) are disinfection byproducts[3] and include the individual chemicals chloroacetic acid, dichloroacetic acid, trichloroacetic acid, bromoacetic acid, and dibromoacetic acid. This group of chemicals is formed in drinking water during disinfection when chlorine reacts with naturally occurring organic material in water sources such as rivers and lakes.

How it's Tested: Chlorination of drinking water is needed to kill harmful organisms such as bacteria and viruses that could cause serious illnesses. However, factors such as the amount of organic matter in water sources, temperature, and the amount of chlorine added affect the amount of HAA5 formed. Specifically, the source of potential HAA5 in Pace's drinking water would come from the chlorination of water provided by the Mount Pleasant Water District. The U.S. Environmental Protection Agency (EPA) sets a maximum contaminant level goal[2] of 0.06 parts per million (equivalent to 0.06 milligrams per liter) for HAA5.

Effect on Health: In regards to health, some studies suggest that drinking chlorinated water containing HAA5 for long periods of time, such as 20 to 30 years, can lead to an increased risk of cancer, low birth weight, miscarriage, birth defects, and other health effects. However, the evidence in these studies is not strong enough to conclude that HAA5s were a major factor contributing to the observed increased risks. Compared to the risks for illness from drinking inadequately disinfected water, the risks for adverse health effects from HAA5s in drinking water are small.

Read more from the NYS Department of Health and the US Environmental Protection Agency (EPA).

## Trihalomethanes (TTHMs)

Trihalomethanes (TTHMs) are disinfection byproducts[3] and include the individual chemicals chloroform, bromoform, bromodichloromethane, and chlorodibromomethane. Disinfection byproducts are formed in drinking water during disinfection when chlorine reacts with naturally occurring organic material in water sources such as rivers and lakes.

How it's Tested: The U.S. Environmental Protection Agency (EPA) sets a maximum contaminant level goal[2] of 0.08 parts per million (equivalent to 0.08 milligrams per liter) for TTHMs. Chlorination of drinking water is needed to kill harmful organisms such as bacteria and viruses that could cause serious illnesses. However, factors such as the amount of organic matter in water sources, temperature, and the amount of chlorine added affect the amount of TTHMs formed. Specifically, the source of potential TTHM in Pace's drinking water would come from the chlorination of water provided by the Mount Pleasant Water District.

Effect on Health: In regards to health, some studies suggest that drinking chlorinated water containing TTHMs for long periods of time, such as 20 to 30 years, can lead to an increased risk of cancer, low birth weight, miscarriage, birth defects, and other health effects. However, the evidence in these studies is not strong enough to conclude that TTHMs were a major factor contributing to the observed increased risks. Compared to the risks for illness from drinking inadequately disinfected water, the risks for adverse health effects from TTHMs in drinking water are small.

Read more from the NYS Department of Health, the US Environmental Protection Agency (EPA), and the CDC.

## Turbidity

Turbidity is the measure of the relative clarity of water. The greater the suspended particles, the higher the level of turbidity and the cloudier the water. Materials that cause water to be turbid include clay, silt, very tiny inorganic and organic matter, algae, dissolved colored organic compounds, and plankton and other microscopic organisms. Some pollutants, such as metals, insoluble toxins, and bacteria, can attach themselves to suspended particles. The particles in cloudy water provide "shelter" for microbes by reducing their exposure to disinfectants. A likely source of contamination that leads to higher turbidity levels is soil runoff.

How it's Tested: Turbidity is measured in Nephelometric Turbidity Units (NTU) by shining a light through a sample of water. The higher the NTU, the more cloudy/opaque the water is. New York State's maximum regulatory contamination level[5] limit is 5 NTU.

Effect on Health: Turbidity readings are a good indicator of water quality and potential pollution. The same conditions that cause high turbidity can promote the regrowth of pathogens, such as bacteria and viruses, leading to waterborne disease outbreaks, which have caused intestinal sickness throughout the United States and the world.

Read more from The United States Geological Survey (USGS).

## CLOSING STATEMENT

Thank you for allowing us to continue to provide your family with quality drinking water this year. In order to maintain a safe and dependable water supply we sometimes need to make improvements that will benefit all of our customers. We ask that all our customers help us protect our water sources, which are the heart of our community. Please call our office if you have questions.

## CONTACT INFORMATION

Public Water System Identification Number (PWSID) ID# 5907678

This report was compiled and prepared by your water system operator:


Allied Pollution Control, Inc.

Water & Wastewater Specialists

1273 Route 311

Patterson, New York 12563

Phone: (845) 878-0007

Fax: (845) 878-2104


https://facilitiesrequest.pace.edu/

https://bluecolab.pace.edu/pacewater/

This report contains important information about your drinking water. Translate it, or speak with someone who understands it.

Este reporte contiene información muy importante sobre el agua que usted toma. Haga que se la traduzcan o hable con alguien que la entienda.

Ce rapport contient des informations importantes sur votre eau potable. Traduisez-le ou parlez en avec quelqu'un qui le comprend bien.

Rapò sa a gen enfòmasyon ki enpòtan anpil sou dlo w'ap bwè a. Fè tradwi-l pou ou, oswa pale ak yon moun ki konprann sa ki ekri ladan-l.

Ten raport zawiera bardzo istotną informacje o twojej wodzie pitnej. Przetłumacz go albo porozmawiaj z kimś kto go rozumie.

В этом материале содержится важная информация относительно вашей питьевой воды. Переведите его или поговорите с кем-нибудь из тех, кто понимает его содержание.

這個報告中包含有關你的飲用水的重要信息 。請將此報告翻譯成你的語言 或者詢問懂得這份報 告的人 。

이 보고셔는 귀하의 식수에 관한 매우 중요한 정보를 포함하고 있습니다. 이 정 보에 대해 이해하는 사람에게 그 정보를 번역하거나 통역해 받으십시오.

*Last year, your tap water met all State drinking water health standards. We are proud to report that our system did not violate a maximum contaminant level or any other water quality standard.*

El año pasado, el agua de su grifo cumplió con todas las normas estatales de salud para el agua potable. Nos enorgullece informar que nuestro sistema no infringió ningún nivel máximo de contaminantes ni ninguna otra norma de calidad del agua.

L'anno scorso, l'acqua del vostro rubinetto ha rispettato tutti gli standard sanitari statali per l'acqua potabile. Siamo orgogliosi di comunicare che il nostro sistema non ha superato i livelli massimi di contaminazione né altri standard di qualità dell'acqua.

No ano passado, a sua água encanada atendeu a todos os padrões estaduais de saúde para água potável. Temos orgulho de informar que nosso sistema não ultrapassou o nível máximo de contaminantes nem violou qualquer outro padrão de qualidade da água.

L'année dernière, l'eau du robinet a respecté toutes les normes sanitaires de l'État en matière d'eau potable. Nous sommes fiers d'annoncer que notre système n'a enfreint aucun niveau maximal de contaminants ni aucune autre norme de qualité de l'eau.